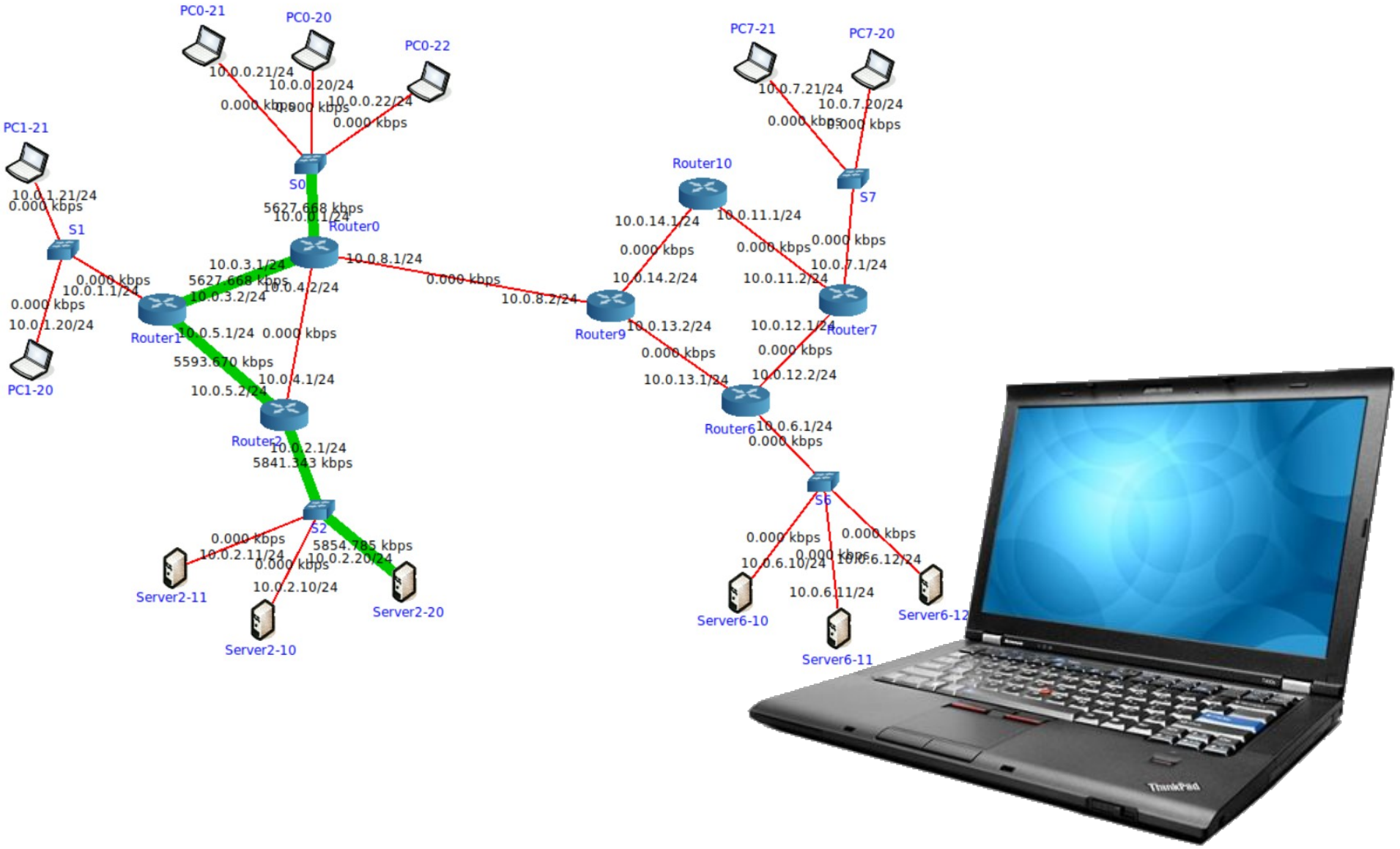
Investigating Linux Network Behaviour Using Open-Source Network Emulators

Brian Linkletter

Network in a laptop



Network Emulation

- Orchestrate the setup and configuration a virtual network scenario in a user-friendly way
- May use Linux containers or VMs, or both, for “routers”
- May use Linux bridging or OVS for “switches”
- Use cases
  - Creating “black box” test beds for external nodes
  - Rapidly set up ad-hoc testing or demonstration scenarios
  - Practice network operations in safe environment
- Many Linux network emulation tools are available
  - Choose based on virtualization, scripting language supported, user interface,

Emulation examples

TCP troubleshooting

- Round trip delays and window sizes
- Bad links(bit errors)

Routing protocol troubleshooting

- Changing network topology
- Silent failures

Delay

No delay configured. Total RTT 0.1 ms

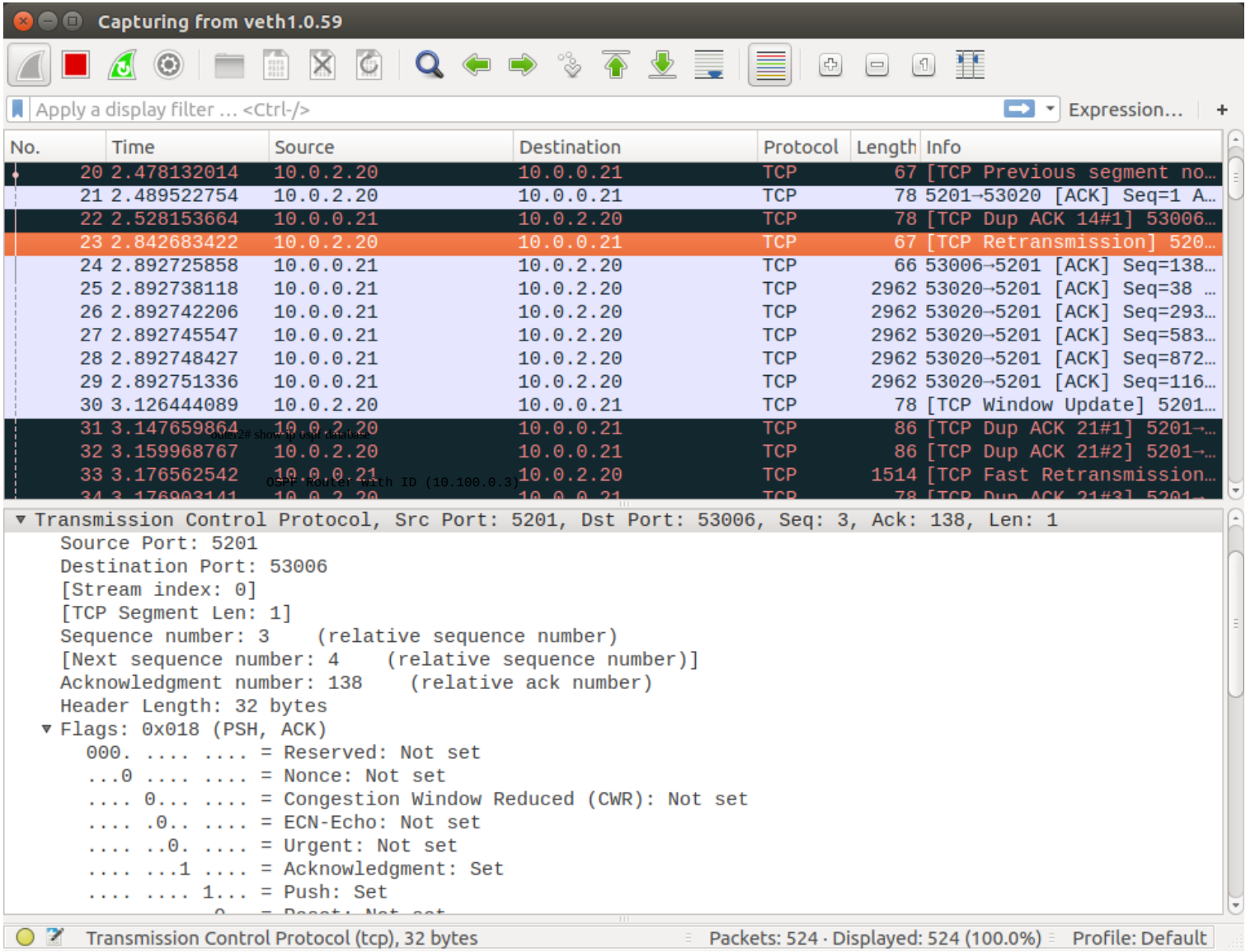
```
<7/PC0-21.conf# iperf3 -c 10.0.2.20 -O 2 -t 5 -w 64K
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-5.00   sec  10.6 GBytes  18.2 Gbits/sec    0  sender
[  4]   0.00-5.00   sec  10.7 GBytes  18.3 Gbits/sec     receiver
```

Delay configured. Total RTT 50 ms

```
<97/PC0-21.conf# iperf3 -c 10.0.2.20 -O 2 -t 5 -w 64K

[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-5.00   sec  5.98 MBytes  10.0 Mbits/sec    4  sender
[  4]   0.00-5.00   sec  6.04 MBytes  10.1 Mbits/sec     receiver
```

Bit errors impact on TCP



Routing failure

Before

```
Router2# show ip ospf database

       OSPF Router with ID (10.100.0.3)

                Router Link States (Area 0.0.0.0)

Link ID         ADV Router      Age  Seq#       CkSum  Link count
10.100.0.1      10.100.0.1       206 0x8000000e 0xea43 5
10.100.0.2      10.100.0.2       811 0x8000000a 0x8fc8 4
10.100.0.3      10.100.0.3       815 0x80000009 0xf260 4
10.100.0.6      10.100.0.6       679 0x80000009 0xf234 4
10.100.0.7      10.100.0.7       680 0x80000009 0x83a4 4
10.100.0.9      10.100.0.9       717 0x8000000a 0xfb05 4
10.100.0.10     10.100.0.10      697 0x80000007 0x1533 3
```

After

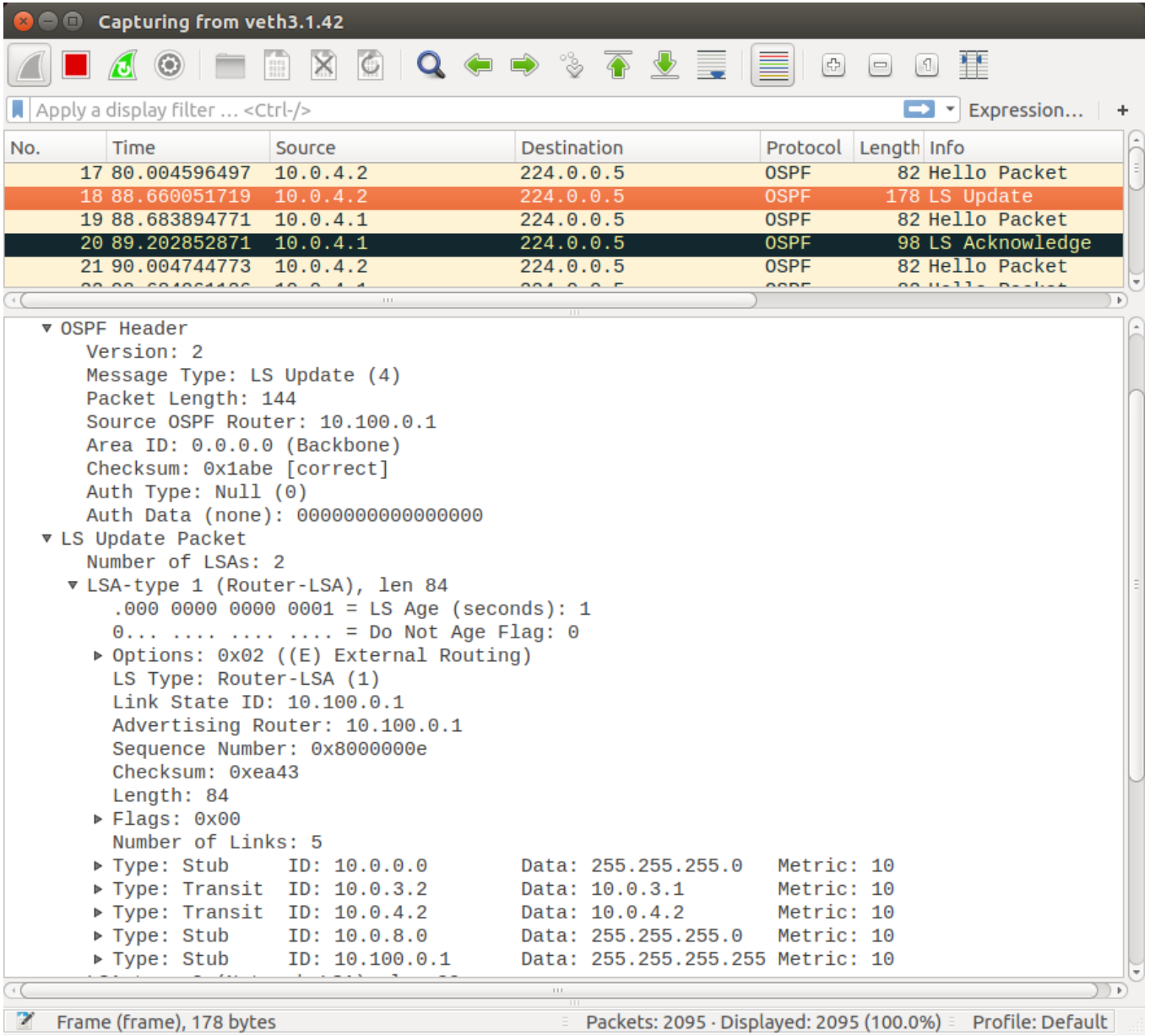
```
Router2# show ip ospf database

       OSPF Router with ID (10.100.0.3)

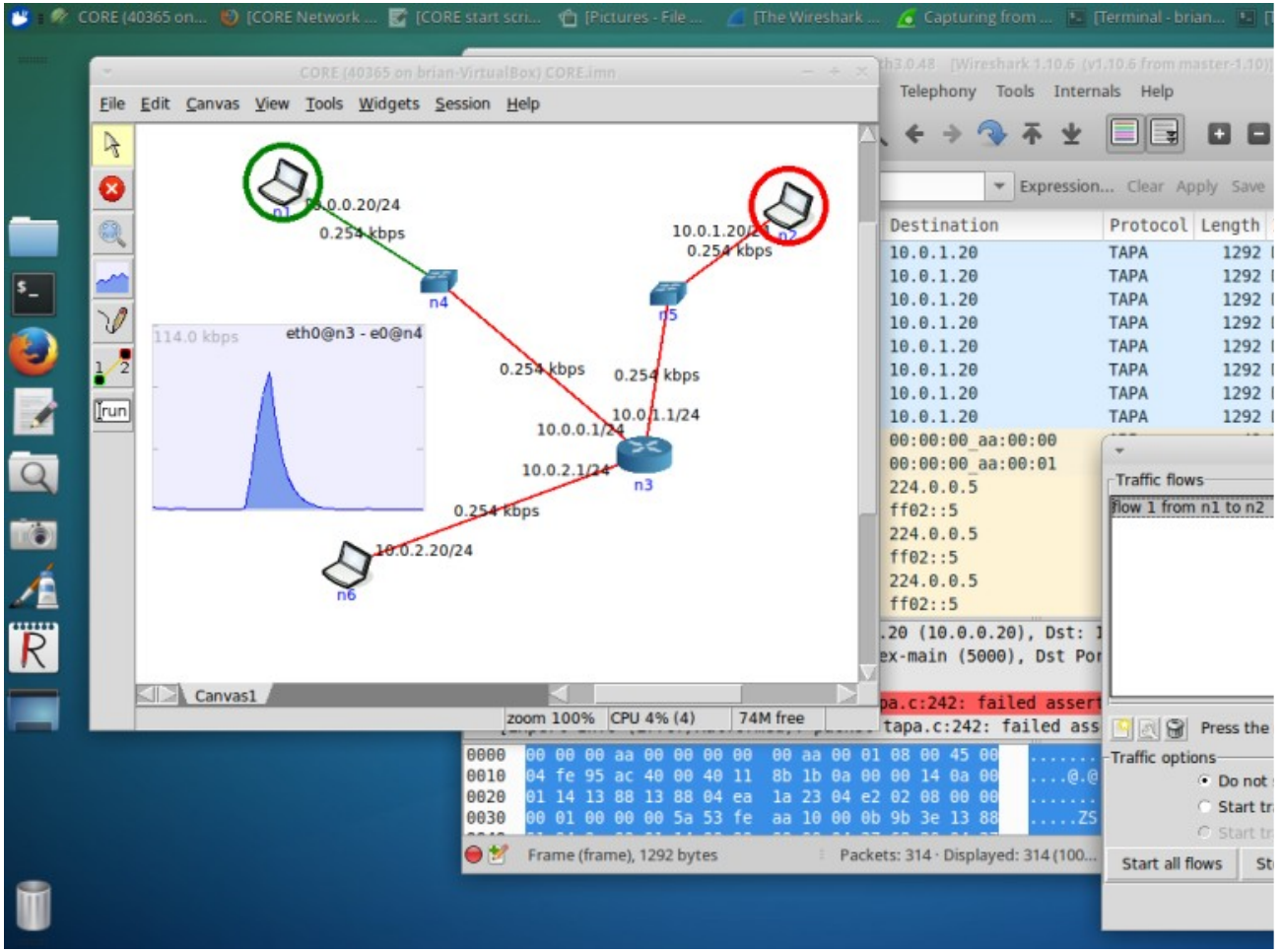
                Router Link States (Area 0.0.0.0)

Link ID         ADV Router      Age  Seq#       CkSum  Link count
10.100.0.1      10.100.0.1      1026 0x80000010 0xe645 5
10.100.0.2      10.100.0.2      1596 0x8000000c 0x8bca 4
10.100.0.3      10.100.0.3      1547 0x8000000b 0xee62 4
```

Routing Failure



CORE



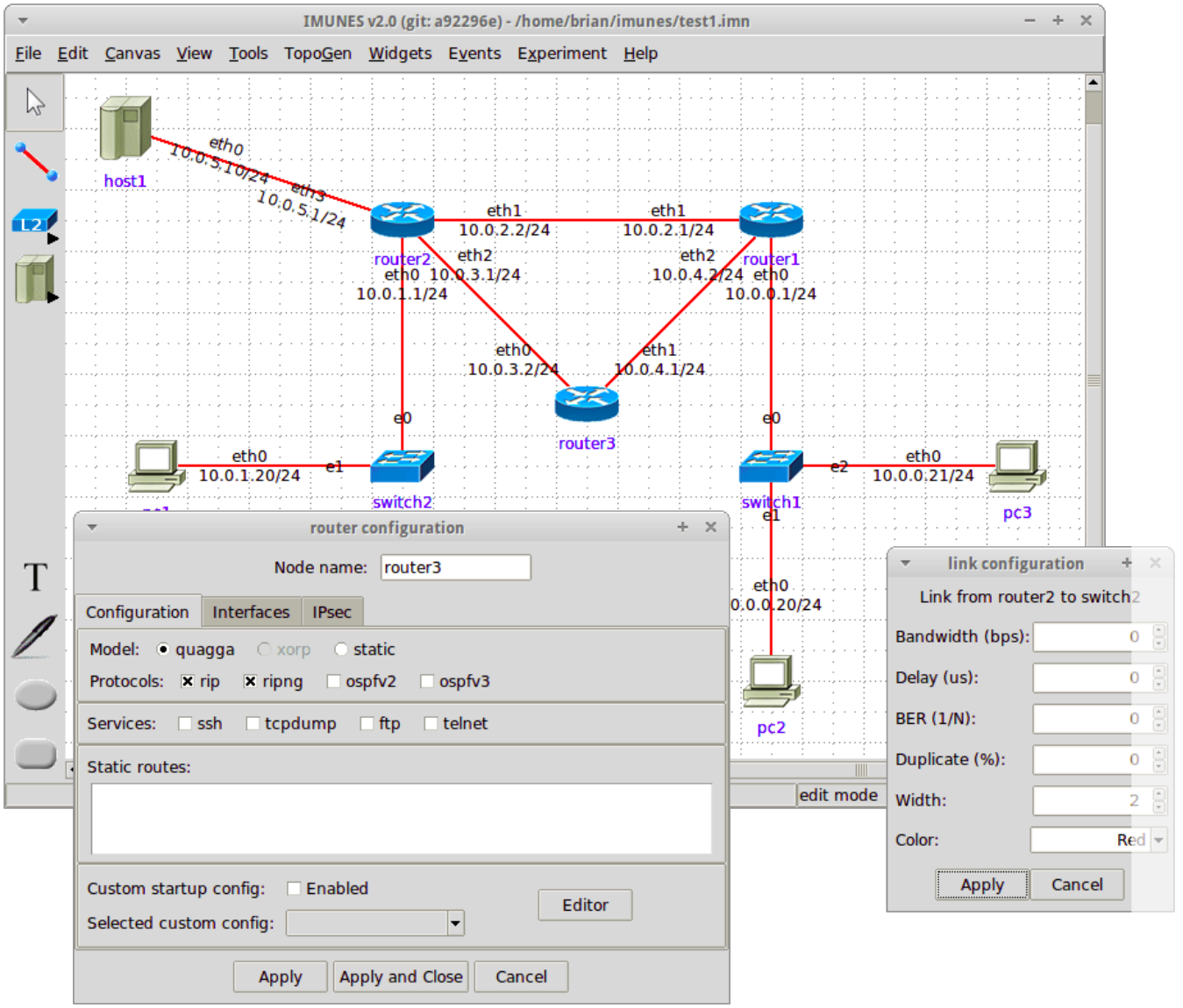
Network namespaces

Linux bridge

Python

XML

IMUNES



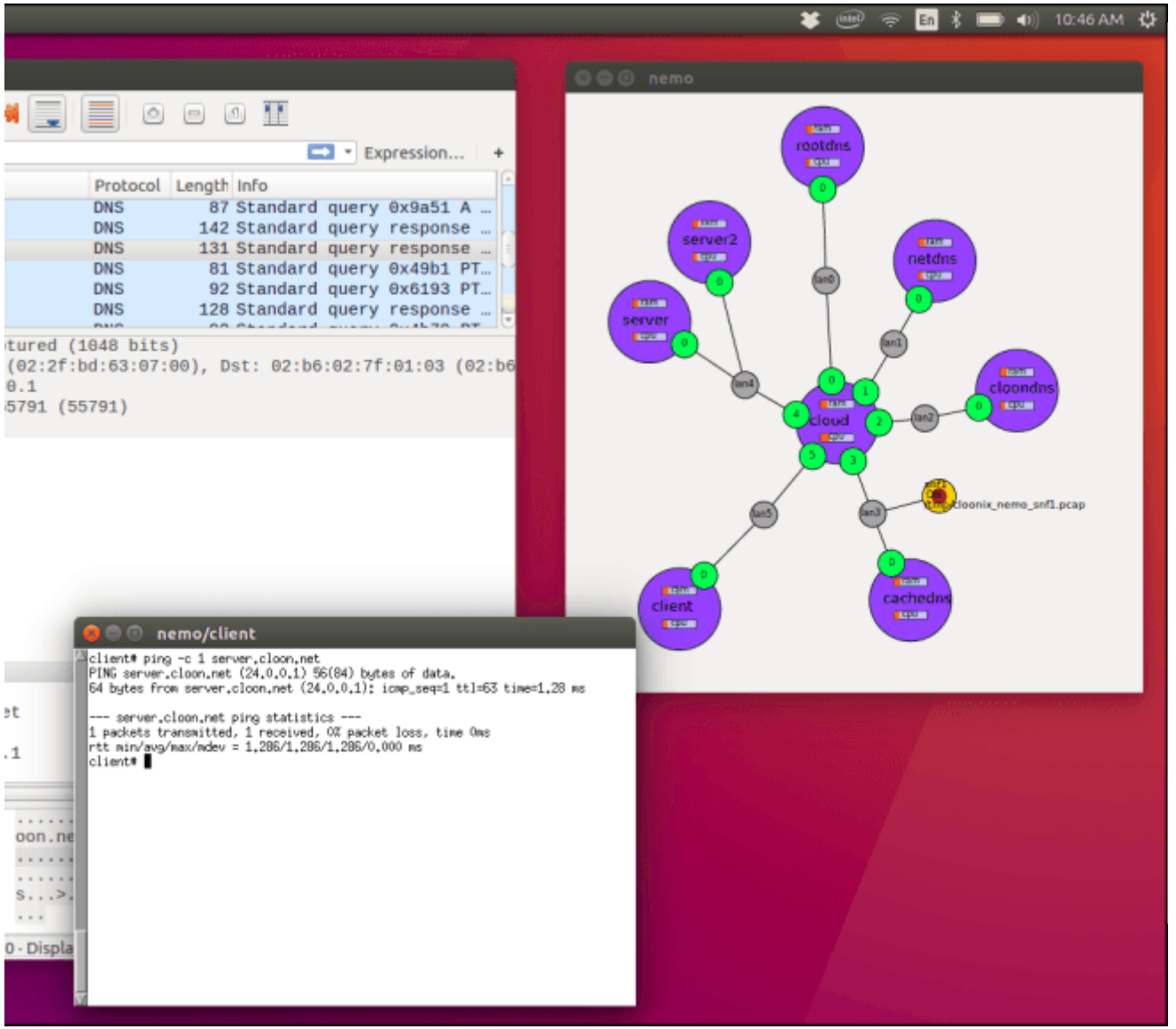
Docker containers

OVS

Python

XML

Cloonix

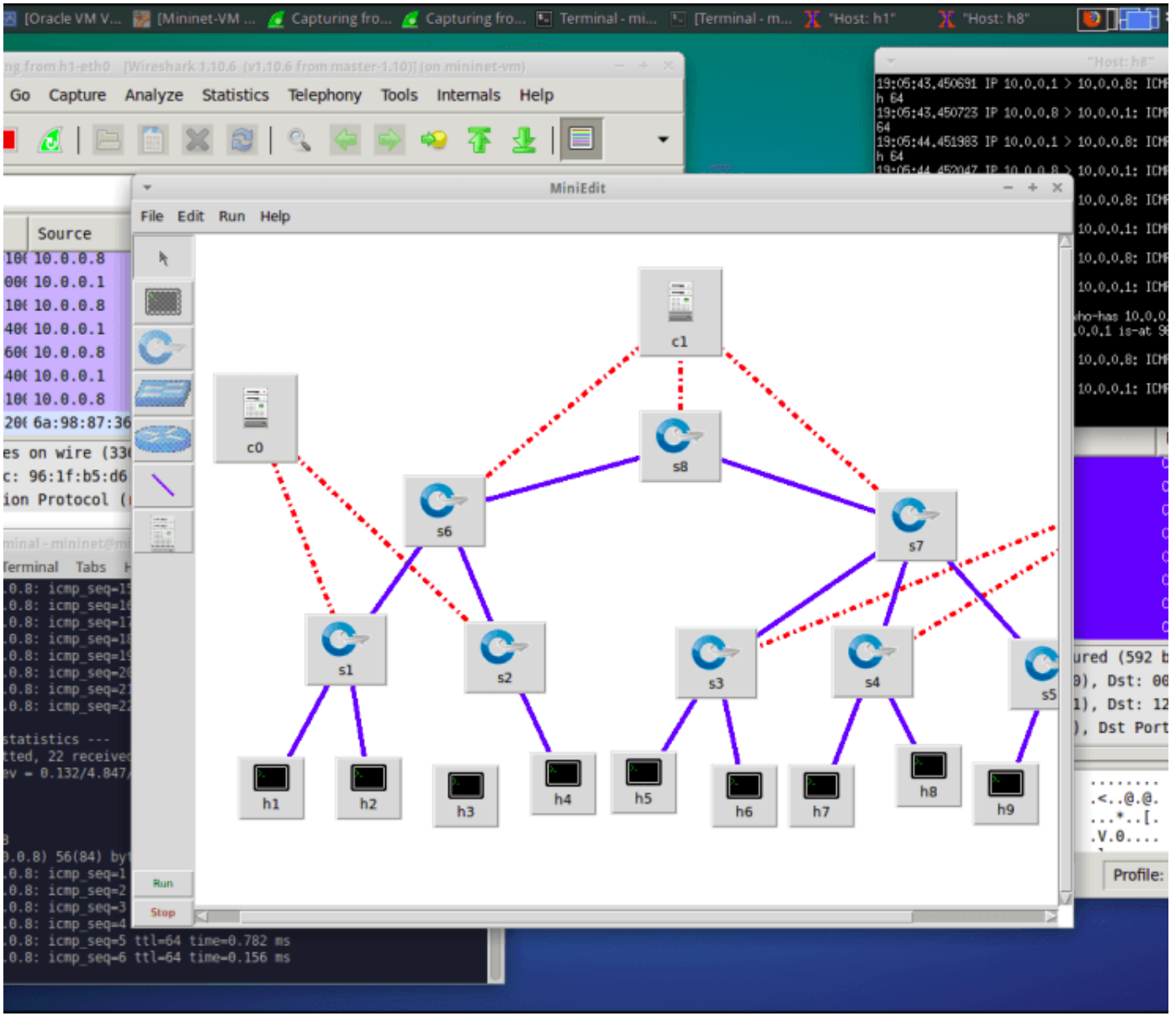


KVM VMs

cloonix switch

Shell scripts

Mininet



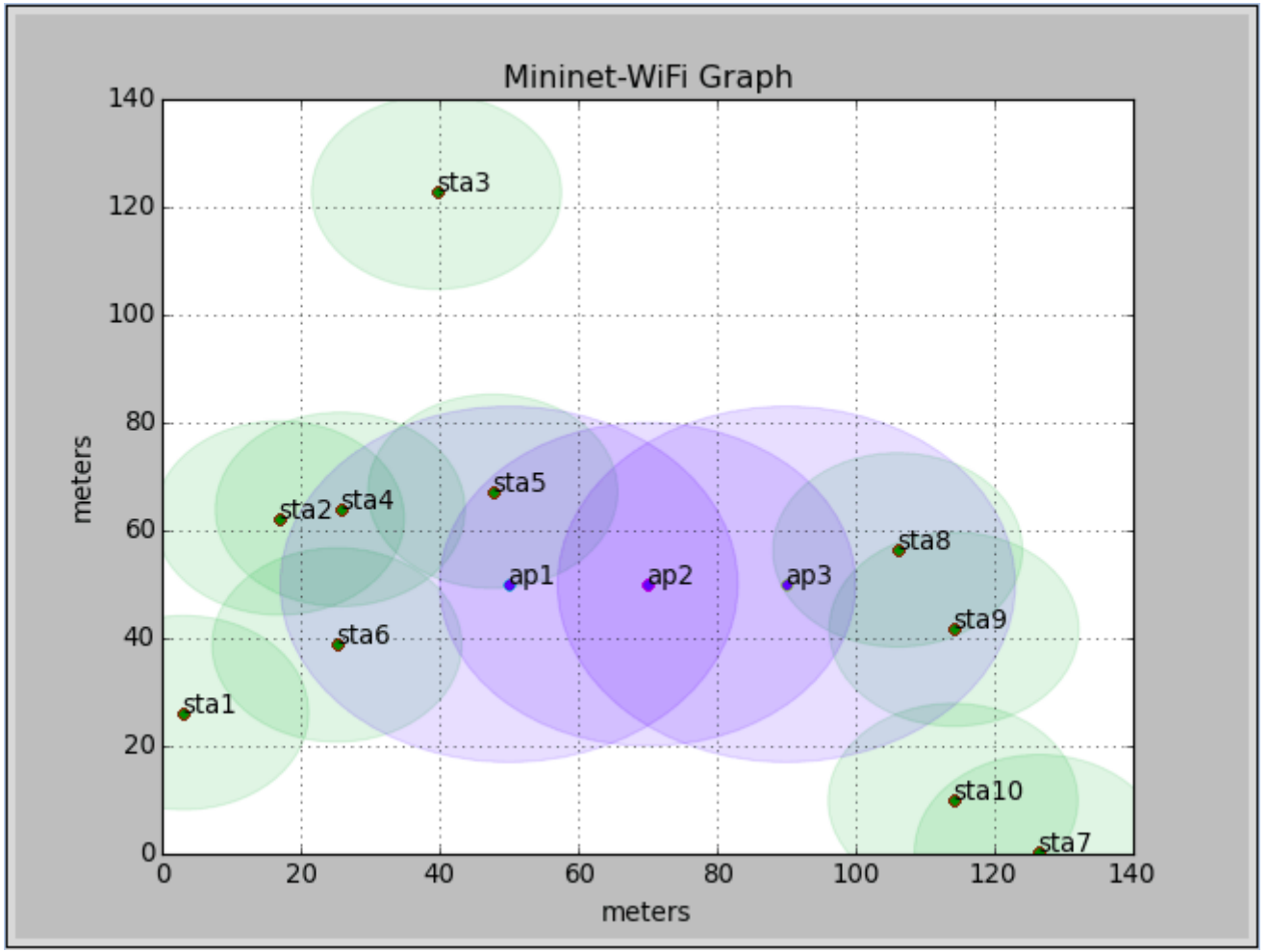
Network namespaces

OVS

Python

XML

Mininet WiFi



WiFi driver

Network namespaces

OVS

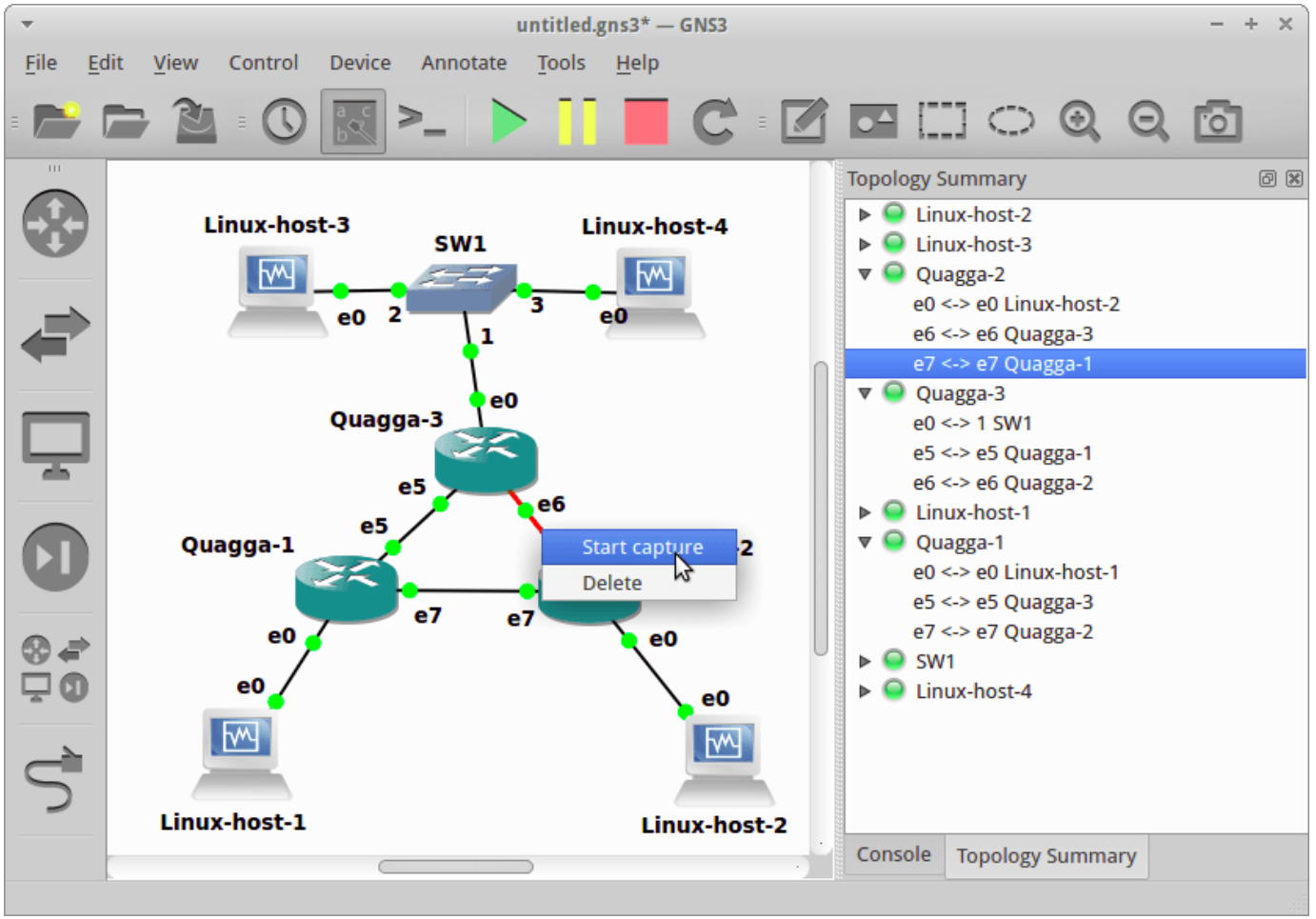
Python

XML

VNX and NetKit

- Both based on CLI user interface
- VNX supports KVM and LXC
- NetKit supports UML
- Each has unique network description language
- Each offers many pre-packaged network emulation scenarios
- Created for University-level networking courses

GNS3

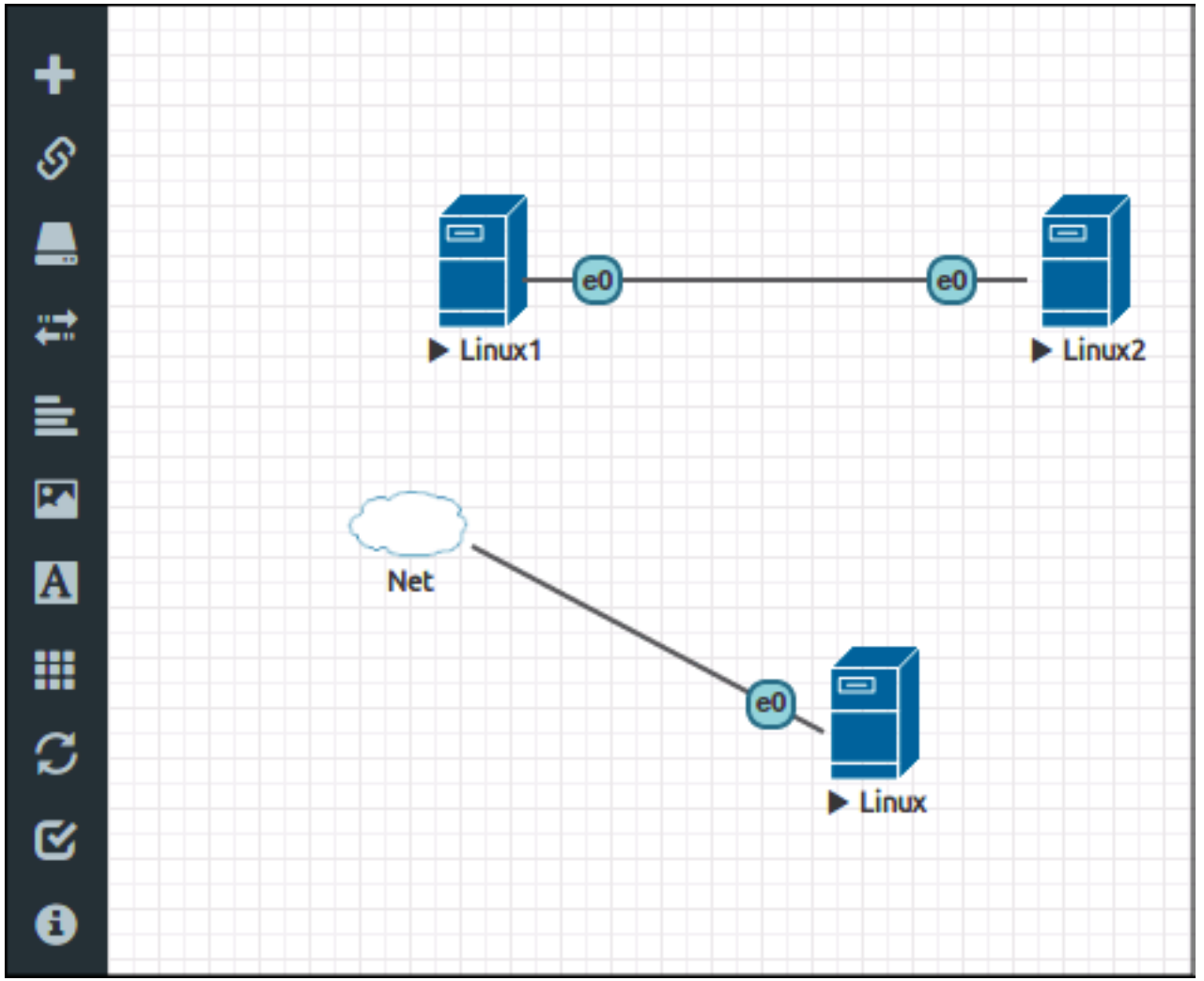


QEMU
VirtualBox
VMware

Dynamips
switching

Becoming more
“Linux friendly”
with new
development

EVE (UNetLab)



QEMU/KVM

custom switch

Linux-friendly

Links

- CORE https://github.com/coreemu/core
- IMUNES https://github.com/imunes/imunes
- Cloonix http://virtual-network-kvm.net/
- Mininet http://mininet.org/
- Mininet-WiFi http://www.ramonfontes.com/mininet-wifi/
- VNX http://www.dit.upm.es/vnx
- Netkit http://wiki.netkit.org
- GNS3 https://www.gns3.com/
- EVE http://www.eve-ng.net/

Other network emulators

- OFNet http://sdninsights.org/
- Shadow https://shadow.github.io/
- Marionnet http://www.marionnet.org/site/index.php/en/
- NetMirage https://crysp.uwaterloo.ca/software/netmirage/
- YANS https://github.com/kennethjiang/YANS

Thank You

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